NAME : DEERAJ S DATE : 08/09/26

SRN : PES2UG25CS806 LAB : 03

DELIVERABLE – 02

Written Justification :

Architecture Selection

We chose Layered Architecture for the Pet Adoption & Shelter Management System.

Architectural Choice and Reasons

Layered Architecture is suitable because it provides a clear separation between the presentation, business logic, and data layers. This makes the system easier to understand, develop, maintain, and modify.

First, it separates different responsibilities into independent components such as User Interface, Authentication & Authorization, Media & Community Management, Event Management, Notification, and Database. This matches the functional requirements of our system, which include managing media, community content, shelter events, and event registrations.

Second, the architecture improves maintainability and scalability because changes in one layer can be made without significantly affecting the other layers. For example, changes to the Event Management component can be handled within the business layer without changing the database or user interface.

Security Advantage

Layered Architecture allows authentication and authorization to be handled separately in the business layer. This helps ensure that only authorized Shelter Staff and Administrators can perform content-management operations, satisfying the security requirement of the system.

Performance Benefit

The separation of business logic and data access allows the system to efficiently retrieve Media, Events, and Community information from the database. This supports the requirement that these pages should load within 3 seconds under normal operating conditions.